

Kavindu Ranathunga

Full-Stack Engineer

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Full-stack engineer with seven months of production experience at Sri Lanka Telecom Mobitel, where I led full-stack delivery on a platform built by a team serving trainees and administrators. Specialize in high-performance interactive UI systems, resilient API architectures, and end-to-end delivery through polished and mobile responsive interfaces. Final-year Software Engineering student at SLIIT.

EXPERIENCE

Sri Lanka Telecom Mobitel -Full-Stack Developer, Intern

Feb – Aug 2026

TalentHub

MERN, Gemini API, Framer Motion

Company-wide platform for running SLT Mobitel's trainee programme, built by a team of 20 engineers.

- Rearchitected the interface across 27 of the platform's 35 pages, spanning the trainee portal and administrative console, unifying dashboards, logbooks, attendance, seat reservation, and leave management into a consistent design system with unique brand identity.
- Engineered a multi-tier logbook-quality assessment feature by routing through local heuristic checks before calling the Google Gemini API, keeping the feature operational during API outages; later optimized to a AI batch validator to stay within daily RPM RPD rate limits.
- Engineered a custom interactive seat-plan module for seat reservation, replacing static seat lists with a pan-and-zoom 96-seat visual layout optimized for touch input on mobile.
- Consolidated duplicated working-day and holiday-calculation logic into one backend module, standardizing how overdue submissions were identified, and wrote a verification script to validate the resulting figures.
- Built a configurable system of seven animated login-screen themes for Sri Lankan and international holidays, with an admin control to select the active theme.
- Diagnosed and resolved cross-platform performance defects, including dashboard load times on mobile, form-field rendering bugs on iOS, and layout failures on small screens.

EasyApply

React, Vite, i18next

- Solo development of EasyApply, converting nine of SLT Mobitel's paper application forms into guided, multi-step online forms covering new connections, relocations, ownership transfers, refunds, and service changes.
- Delivered the complete interface in English, Sinhala, and Tamil for company-wide accessibility.

PROJECTS

SchemaShift

Next.js, TypeScript, Supabase, PLpgSQL

github.com/kavindu-rakn/SchemaShift

- Architected a schema-migration tool for catalogue data that runs live impact analysis before any change is committed, surfacing affected categories, record counts, and unpreservable values.
- Implemented schema-resolution logic as a PostgreSQL function mirrored in TypeScript, enabling instant client-side change previews with no server round-trip.
- Enforced strict read-only guarantees on impact analysis, validated by a test suite comparing full record state across 18 consecutive analyses.
- Applied migrations within a single atomic transaction with version-based rollback; validated transactional integrity with a test that injects a mid-migration failure and confirms no partial writes persist. Covered by 234 automated tests.

Hotel Tamarind Tree

Next.js, Prisma, PostgreSQL, Auth.js, Vercel, Resend

github.com/kavindu-rakn/HotelTamarindTree

- Built a hotel booking platform end-to-end from a written specification across five independently planned phases, including a three-stage booking flow backed by live availability checks and an admin dashboard for bookings, rooms, and enquiries.
- Implemented availability logic to prevent double-booking of the same room, paired with a scheduled nightly job that automatically returns abandoned reservations to inventory.

Horologia

Three.js, GSAP, Web Audio API, Firebase

github.com/kavindu-rakn/Horologia

- Built a real-time interactive 3D visualization of a mechanical watch movement that disassembles as the page scrolls, rendering 14 procedurally generated components including the gear train, escapement, balance wheel, and tourbillon.
- Synthesized mechanical ticking audio at 28,800 beats per hour via the Web Audio API, with per-component material inspection and save-to-gallery sharing through Firebase.

Other projects:

- Prompta - AI prompt manager with folders, sharing, and cross-LLM shortcuts (Next.js, Supabase, GoogleAuth)
- Luna - Real-time lunar-cycle visualizer with a draggable 3D model (React, Three.js, SunCalc, GSAP, Drei)

SKILLS

Core Languages

JavaScript, TypeScript, SQL, HTML, CSS, Python, Java

Frontend & Graphics Architecture

React, Next.js, Vite, Tailwind, Framer Motion, GSAP, Three.js, Chart.js, Leaflet, i18next

Backend & Distributed Systems

Node, Express, Server Actions, REST APIs, JWT, GoogleAuth, Nodemailer, Resend

Databases & ORMs

PostgreSQL, Supabase (RLS, plpgsql, triggers), Prisma, MongoDB, Mongoose

AI & Web APIs

Google Gemini API, Claude Code, TensorFlow.js, face-api.js, Web Audio API

Tooling & DevOps

Git & PR workflows, GitHub Actions, Vercel, Firebase, automated testing, PDF/spreadsheet generation, localization (EN/SI/TA)

EDUCATION

• Sri Lanka Institute of Information Technology, Colombo

BSc (Hons) Information Technology, specializing in Software Engineering

• Royal College, Colombo

Biological and Physical Sciences